

Skills Summary

Solving Exponential and Logarithmic Equations

Use this reference while completing your worksheet or when studying.

One sentence, two spellings: $\log_b M = k$ and $b^k = M$ mean the same thing. Every technique below is that swap, used in one direction or the other.

The three laws: $\log_b(MN) = \log_b M + \log_b N$, $\log_b\left(\frac{M}{N}\right) = \log_b M - \log_b N$, $\log_b(M^k) = k \log_b M$.

Change of base: $\log_b M = \frac{\log M}{\log b}$, which is how a calculator produces a decimal for any base.

1. Unknown in the exponent

If both sides can be written with the same base, match the exponents — exact and no calculator. If they cannot, take a logarithm of both sides and use the power law to bring the exponent down in front.

Isolate the exponential factor first: a coefficient in front is not part of the power.

Example: Solve $3^x = 20$ to the nearest thousandth.

Step 1. Twenty is not a power of three, so the bases cannot be matched — take a logarithm of both sides instead.

Step 2. $x \log 3 = \log 20$, so $x = \frac{\log 20}{\log 3}$, and evaluating gives 2.72683...

$$x \approx \mathbf{2.727}$$

Try it: Solve $5^x = 90$ to the nearest thousandth.

check: 962.7

2. Unknown inside a logarithm

Get the equation down to one logarithm equal to a number, then rewrite it in exponential form: $\log_b x = k$ becomes $x = b^k$.

Because a logarithm needs a positive argument, test every candidate root in the original equation — condensing can invent roots the equation never had.

Example: Solve $\log_4 x = 3$.

Step 1. One logarithm already sits alone on the left, so the exponential rewrite finishes the problem in a single move.

Step 2. $\log_4 x = 3$ means $x = 4^3$.

$$x = \mathbf{64}$$

Try it: Solve $\log_3 x = 4$.

check: 18

3. Rewriting with the logarithm laws

Expanding turns one logarithm into several; condensing does the reverse and is what makes an equation solvable, because only a single logarithm can be rewritten in exponential form. Clear coefficients with the power law first, then use the product law for a sum and the quotient law for a difference.

Example: Expand $\log_5(25x^3)$ for $x > 0$.

Step 1. The argument is a product with a power inside it, so the product law splits it and the power law then clears the exponent.

Step 2. $\log_5 25 + \log_5(x^3) = \log_5 25 + 3 \log_5 x$, and $\log_5 25 = 2$ because $5^2 = 25$.
 $2 + 3 \log_5 x$

Try it: Condense $3 \log_2 x + \log_2 7$ into a single logarithm.

(x_2)² $\log_2$:xpep

(!) **Watch out:** $\log(M+N)$ is not $\log M + \log N$ — the laws apply to products and quotients inside the logarithm, never to sums. And $\frac{\log 5}{\log 2}$ is not $\frac{5}{2}$: take the logarithms first, divide second.